

EDUCATION

EMORY UNIVERSITY

Master of Public Health in Biostatistics

Atlanta, GA
Current

- Relevant Coursework: Statistics, Machine Learning

NC State University

Bachelor of Science in Statistics; Minor in Economics

Raleigh, NC
08/2019-12/2023

- Honors: Dean's Honors List (4 Semesters)

EXPERIENCE

Mental Health Study

Atlanta, GA
Current

Quantitative Research Assistant

- Extracted, queried, and aggregated complex and disparate datasets (over 80,000 records) from diverse electronic health databases using SQL to derive key analytical insights.
- Developed multivariate statistical models using R to analyze complex relationships, employing advanced data imputation techniques to clean datasets with 20% missing values and ensure high data quality.
- Partnered closely with a cross-functional, multidisciplinary team to interpret complex statistical findings and visualize data trends, successfully communicating insights to non-technical stakeholders for internal strategic reporting.

Chinese Academy of Sciences

Nanjing, China
01/2024-06/2024

Research Assistant, Institute of Soil Science

- Built spatial segmentation and clustering models using SAS to identify critical contamination patterns, providing quantitative analysis to solve complex regional problems.
- Managed rigorous data normalization and quality control protocols to ensure 100% data integrity for advanced statistical modeling.

Shanxi Securities Research Institute

Shanghai, China
07/2022-08/2022

Industry Intern

- Integrated and analyzed large-scale financial data from multiple disparate sources (Wind, Choice) to evaluate business performance and support data-driven business decisions.
- Engineered automated analytical scripts in Python and R to calculate key valuation metrics, reducing manual processing time and significantly enhancing pipeline efficiency.
- Performed business intelligence tasks by synthesizing findings into data-driven Excel presentations, answering complex questions, and earning the trust of senior stakeholders.

ACADEMIC PROJECTS

SAS Projects

- Clinical Trials Case Study: Performed data manipulation and cleaning on complex clinical trial datasets using proc SQL. Analyzed patient data to derive conclusions about clinical efficacy and identify characteristics that influence patient outcomes.
- Prospective Evaluation of Radial Keratotomy (PERK) Group Project: Conducted t-tests and ANOVA to find relationships between patient characteristics and surgical outcomes. Produced numerical and graphical summaries to report on the effectiveness of a surgical procedure, showing significant improvement post-surgery.

R Projects

- FAERS Project: Cleaned and merged seven complex and heterogeneous datasets from the FDA Adverse Event Reporting System to create a unified data pipeline for machine learning application. Built, evaluated, and performed gap analysis on comparative machine learning classification models (LDA, QDA, KNN) to predict outcomes, achieving a prediction accuracy of 80%.

SKILLS

- Data & Programming:** Python (Pandas, NumPy), R (ggplot2, dplyr), SAS (Base, Macro, SQL), SQL (MySQL), Java.
- Analytics & Statistics:** Hypothesis Testing, Regression Analysis, Machine Learning (Classification, Clustering), A/B Testing, Data Cleaning.
- Business Intelligence & Tools:** Microsoft Excel, PowerPoint, Tableau.
- Languages:** Mandarin (Native), English (Professional Working Proficiency).